HOMEWORK 3 (FOR THE 3RD AND 4TH WEEK)

Armstrong's book Chapter 3, Exercise 6, 8, 9, 10, 11, 13, 14, 21, 25, 26 together with the following exercise. Note that in exercise 25, a map means a continuous map. In exercise 26, an open (closed) map means it maps open (closed) sets to open (closed) sets.

Exercise 1. *In the class, we proved that a subset in $\mathbb{R}^n$ is compact if and only if it is closed and bounded.*

(1) *Give an example to show that a closed and bounded subset in a metric space is not necessarily compact in general.*

(2) *Let X be a metric space and let $\epsilon > 0$. A subset $E \subset X$ is called an $\epsilon-$**grid** if for any $x \in X$ there exists $e \in E$ such that $d(e, x) < \epsilon$. A metric space is called **totally bounded** if for any $\epsilon > 0$ there exists a finite $\epsilon-$grid. Prove that a subset in a complete metric space is compact if and only if it is closed and totally bounded.*